

The Research Problem, Questions and Objectives

90 minutes · 5 self-check questions

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How to move from a broad interest to a question that is specific enough to answer, and how aims, objectives, hypotheses and conceptual frameworks fit together.

Learning objectives

- Write a problem statement that establishes magnitude, consequence and gap
- Convert a broad topic into a focused, answerable research question
- Apply FINER and PICO to test whether a question is worth asking
- Distinguish aims, objectives, research questions and hypotheses
- Draw a conceptual framework linking your key variables or concepts

The problem statement

- A problem statement answers four things in order: what is the problem, how big is it, what follows if it is not addressed, and what is not yet known
- Magnitude needs evidence — prevalence, incidence, cost, or documented reports, each cited
- The gap is the pivot of the whole proposal: it must be a gap in knowledge, not merely a gap in your own reading
- A gap can be empirical (nobody has measured this), methodological (measured badly), theoretical (no adequate explanation) or contextual (established elsewhere but never tested here)
- Contextual gaps are legitimate and are often the strongest basis for research in Ghana and across Africa

IN PRACTICE

"No study has been done in my district" is only a gap if there is reason to expect the local answer might differ. Say why it might — different health system, different exposure, different population structure.

From topic to question

- Narrow along four axes: population, concept or exposure, outcome, and setting or time
- "Malaria in children" is a topic; "Does distance to a health facility predict delayed treatment for malaria in under-fives in the Upper East Region?" is a question
- A good question is answerable with the data and time you actually have

- If you cannot name the analysis that would answer it, the question is not yet specific enough
- Quantitative questions specify variables and their relationship; qualitative questions specify experience, process or meaning

Testing the question: FINER and PICO

- FINER — Feasible, Interesting, Novel, Ethical, Relevant
- Feasibility is the most commonly overlooked: participants, time, funding, skills and access
- PICO structures comparative quantitative questions: Population, Intervention or exposure, Comparison, Outcome
- Add T for timeframe (PICOT) when duration matters
- For qualitative work, SPIDER is a better fit: Sample, Phenomenon of Interest, Design, Evaluation, Research type

FRAMING THE SAME TOPIC THREE WAYS

Element	Quantitative (PICO)	Qualitative (SPIDER)
Who	Under-fives with confirmed malaria	Caregivers of under-fives treated for malaria
What	Distance greater than 5 km to facility	Experience of seeking and delaying care
Compared with	Distance of 5 km or less	Not applicable
Outcome	Treatment delayed beyond 24 hours	How caregivers account for delay
Analysis implied	Logistic regression, odds ratio	Thematic analysis of interviews

Aims, objectives and hypotheses

- The aim is one sentence stating the overall purpose — usually beginning "To determine...", "To explore...", "To evaluate..."
- Objectives break the aim into specific, ordered, measurable steps, each starting with an action verb
- Three to five objectives is usual; more than six normally signals an unfocused study
- A hypothesis is a testable statement of an expected relationship, stated for quantitative work only
- The null hypothesis states no difference or no association; the alternative states the expected effect
- Qualitative studies state research questions rather than hypotheses, because they do not set out to test a prediction

IN PRACTICE

Objectives must map one-to-one onto your analysis plan and, later, onto your results section. If an objective produces no result, or a result answers no objective, something has gone wrong.

Conceptual and theoretical frameworks

- A theoretical framework borrows an existing theory (Health Belief Model, Theory of Planned Behaviour, Social Cognitive Theory) to explain your phenomenon
- A conceptual framework is your own diagram of how the variables in your study relate
- It should distinguish independent, dependent, mediating, moderating and confounding variables
- Drawing it early exposes variables you would otherwise forget to measure
- The framework is also what tells you which variables must be adjusted for in analysis

Worked examples

Narrowing a topic until it can actually be answered

A student arrives with the topic "mental health of students in Ghana". It is a real problem, but as written it cannot be studied.

1. Population: which students? Settle on final-year nursing students at one university.
2. Exposure or concept: what about them? Choose academic workload, measured in timetabled contact hours.
3. Outcome: what will you measure? Depressive symptoms on the PHQ-9.
4. Setting and time: one campus, one semester.
5. The question becomes: "Is timetabled workload associated with PHQ-9 score among final-year nursing students at [University] in the 2026 second semester?"
6. Now name the analysis: linear regression of PHQ-9 score on contact hours, adjusted for age, sex and financial stress. If you cannot name it, the question is still too vague.

THE POINT

The test of a finished question is that it dictates the analysis. If two people would analyse your data differently, the question is not specific yet.

An objective that produces no result

A proposal lists four objectives. The third is "to explore the perceptions of health workers". The study is a cross-sectional survey with a closed-response questionnaire.

1. "Explore perceptions" implies qualitative work — interviews or focus groups.
2. The instrument collects no free text, so nothing in the data can answer this objective.
3. At write-up the student will either drop the objective, or pad the results with a frequency table that does not address it.
4. Either outcome invites the examiner question: why did you state an objective you never met?
5. The fix at proposal stage is cheap: rewrite it as "to describe health workers' reported barriers", or add a qualitative strand and design for it.

THE POINT

Map every objective to the specific analysis that will answer it before submitting the proposal. An unanswerable objective costs nothing to fix now and a chapter of trouble later.

Key terms

Research gap

A specific limitation in existing knowledge that the proposed study will address.

Aim

A single statement of the overall purpose of the study.

Objective

A specific, measurable step that contributes to achieving the aim.

Null hypothesis

The testable statement that no difference or association exists in the population.

Conceptual framework

A diagram or model of the presumed relationships among the study variables.

Confounder

A variable associated with both exposure and outcome that can produce a spurious association.

Activity

Write a one-paragraph problem statement and one research question for your own study. Exchange with a partner, who must identify the gap type and name the analysis your question implies. If they cannot, revise it.

Further reading

- **Hulley et al. — Designing Clinical Research (Ch. 2)** — Origin of the FINER criteria; concise and practical.
- **Cooke, Smith & Booth (2012) — "Beyond PICO: The SPIDER Tool"** — The qualitative counterpart to PICO.
- **Ravitch & Riggan — Reason & Rigor** — The best treatment of conceptual frameworks as an argument rather than a diagram.

Self-check questions

1. Which of the following is a research question rather than a topic?
 - a. Adolescent mental health in Ghana
 - b. The effect of social media on young people
 - c. Is daily social media use of more than three hours associated with depressive symptoms among senior high school students in Accra?
 - d. Investigating depression and technology
2. In FINER, the criterion most often underestimated by postgraduate students is:
 - a. Interesting
 - b. Novel
 - c. Feasible
 - d. Relevant
3. Which statement is a correctly framed null hypothesis?
 - a. Distance to a health facility increases treatment delay.
 - b. There is no association between distance to a health facility and treatment delay.
 - c. Does distance to a health facility affect treatment delay?
 - d. Treatment delay is an important problem in rural districts.
4. A study on a topic well established in Europe but never examined in West Africa is justified by which type of gap?
 - a. Methodological gap
 - b. Theoretical gap
 - c. Contextual gap
 - d. No gap exists
5. The relationship between objectives and the results section should be.

- a. Independent — results may cover whatever the data show
 - b. One-to-one — each objective yields a result and each result answers an objective
 - c. Objectives are for the proposal only and need not appear later
 - d. Results should exceed the objectives to show thoroughness
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Answers

1. **c.** Only option C specifies population, exposure, outcome and setting, and it implies a definite analysis. The others name areas of interest without stating what would be measured or compared.
 2. **c.** Feasibility — access to participants, time, funding and skills — is where most over-ambitious proposals fail. A question can be novel, ethical and relevant and still be impossible within a one-year programme.
 3. **b.** The null hypothesis asserts the absence of an association. Option A is the alternative hypothesis, C is a question, and D is a background claim.
 4. **c.** This is a contextual gap. It is legitimate, but the proposal must explain why the local answer could plausibly differ — different health system, exposure profile or population structure — rather than merely noting absence.
 5. **b.** Objectives are the skeleton of the study. An objective with no result suggests work not done; a result answering no objective suggests analysis that drifted, which invites accusations of fishing for findings.
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